

Anas Sharafeldin Mohamed Osman

Seismogeodesist & Geomatics Engineer — Earthquake cycle geodesy, InSAR, GNSS, geoid modelling

 ORCID ·  LinkedIn ·  ResearchGate ·  Google Scholar

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Education

- 2023 – 2027 (expected) **PhD, Geodesy and Surveying Engineering**, University of Chinese Academy of Sciences, Beijing, China
Institute of Geodesy and Geophysics, CAS
Research: Seismology and Geodesy, InSAR fault slip modelling, seismotectonics
- 2020 – 2022 **MEng, Geomatics Engineering (Cum Laude)**, Institut Teknologi Sepuluh Nopember (ITS), Surabaya, Indonesia
Advisor: Prof Ira Mutiara Anjasmara
Thesis: Gravimetric geoid modelling for Sudan using least-squares collocation; CGPA 3.86/4
- 2012 – 2017 **BSc (Honours), Surveying Engineering**, Omdurman Islamic University, Khartoum, Sudan
Second Class Division One; Academic Excellence Award for highest GPA in cohort

Research Interests

Earthquake cycle geodesy (interseismic, coseismic, postseismic deformation) · InSAR time-series analysis and fault slip modelling · GNSS geodesy and precise positioning · Gravimetric geoid determination and evaluation of global geopotential models · Land subsidence monitoring with multi-temporal InSAR · Geodetic reference frames and satellite geodesy

Research Experience

- Since 2023 **PhD Researcher**, University of Chinese Academy of Sciences / Institute of Geodesy and Geophysics, CAS (*Advisors: Prof Heping Sun, Prof Rumeng Guo*)
- Investigating the earthquake cycle (interseismic, coseismic, postseismic) along the Kamchatka subduction zone using InSAR (Sentinel-1, ALOS-2) and GNSS time series.
 - Developing fault slip models constrained by geodetic observations to understand strain accumulation and release.
- 2020 – 2022 **MSc Researcher**, Institut Teknologi Sepuluh Nopember (*Advisor: Prof Ira Mutiara Anjasmara*)
- Determined a new regional gravimetric geoid model for Sudan using least-squares collocation (LSC).
 - Evaluated high-degree global geopotential models (EGM2008, EIGEN-6C4, GECO) against GPS/levelling benchmarks.
- 2017 – 2019 **Consultant Surveying Engineer**, Dar Consult Engineering Consultants, Khartoum, Sudan
- Led surveying teams for major road and bridge construction projects; performed topographic surveys, setting out, and as-built documentation.
 - Implemented GPS-based techniques improving efficiency by 40%; trained junior surveyors.

Research Publications

Full record on [ResearchGate](#) and [Google Scholar](#).

- Interseismic, Coseismic, and Early Postseismic Slip Associated With the 2025 Mw 8.8 Kamchatka Earthquake**
Yang Xiao, Jingqi Wang, Yijun Zhang, Wenting Zhang, Jiangcun Zhou, Heping Sun, Yabo Zhao, **Anas Osman**, Rumeng Guo
Geophysical Research Letters, May 2026
DOI: <https://doi.org/10.1029/2025gl121473>
- Multi-Temporal Identification and Analysis of Land Surface Deformation with Land Use Land Cover in Makassar City**
Suharman Hamzah, Regita Faridatunisa Wijayanti, Ira Mutiara Anjasmara, Laode M Sabri, **Anas Osman**
Indonesian Journal of Urban and Environmental Technology, Feb 2026
DOI: <https://doi.org/10.25105/urbanenvirotech.v9i1.23368>
- Regional evaluation of global geopotential models and digital elevation models with ground-based gravity and GNSS/levelling data over Sudan**
Anas Osman, Basem Elsaka, Ira Mutiara Anjasmara
Journal of Applied Geodesy, Jan 2025
DOI: <https://doi.org/10.1515/jag-2024-0006>
- Determination of a New Gravimetric Geoid Modelling for Sudan Using the Least-Squares Collocation Technique**
Osman, A.S.M., Anjasmara, I.M.
IOP Conference Series: Earth and Environmental Science, Vol. 1127, 012014
DOI: <https://doi.org/10.1088/1755-1315/1127/1/012014>

1 Accuracy Investigation of the Three-Point Resection Method Through the Distribution of Control Points Across Four Quadrants

Anas Osman, Abdalla M A Mabrouk, Abdalraheem M A Mahjoub, Eltayeb A M Elkhailifa, A Abbas Elhag

Journal of Marine-Earth Science and Technology, Nov 2021

DOI: <https://doi.org/10.12962/j27745449.v2i2.101>

1 Assessment of High-Degree Reference Models and Recent GOCE/GRACE Global Geopotential Models over Sudan Based on GPS/Leveling Data

Osman, A.S.M., Anjasmara, I.M., Ruby, A., Udama, Z.A.

IOP Conference Series: Earth and Environmental Science, Vol. 936, 012035

DOI: <https://doi.org/10.1088/1755-1315/936/1/012035>

Selected Presentations & Workshops

- Nov 2025 **COMET InSAR Training Workshop** (UK NERC) – participant, presented poster on Kamchatka interseismic deformation.
- Oct 2024 **GGOS Days 2024** (Global Geodetic Observing System / IAG) – presented on evaluation of global geopotential models over Sudan.
- Jun 2024 **EarthScope Seismology Skill Building Workshop** – attended and contributed to group discussion on GNSS time series analysis.
- Apr 2024 **Advanced InSAR for Civil Engineering** (AGS, Hong Kong) – participant.
- Mar 2024 **NOAA NGS Webinar Series** – “Evaluating Noise in GNSS Time Series”, “Space Weather Impacts on GNSS”.

Teaching & Supervision

- 2022 – Present **Adjunct Lecturer**, Omdurman Islamic University, Faculty of Engineering Sciences
- Teach undergraduate and graduate courses: Geoinformatics, Geoid Determination, Satellite Geodesy, Advanced Surveying Engineering.
 - Supervise 15+ undergraduate final year projects; develop modern course materials; serve on examination committees.
- 2018 – 2019 **Teaching Assistant (part-time)**, Omdurman Islamic University
- Assisted in surveying practicals and introductory geodesy courses.

Academic Service & Peer Review

- 2025 **Certified Peer Reviewer** – University of Nottingham (Certificate of Publication Peer Review).
- 2024 **Open Science Certificate** – University of Nottingham (promoting reproducible research).

Technical Skills

Geodetic Software	SNAP, GAMMA, ROI_PAC, BERNESE GNSS, GIPSY-OASIS, GRAVSOFT
Programming & Analysis	MATLAB, PYTHON, R, FORTRAN, GMT, LATEX
Engineering & GIS	AUTOCAD CIVIL 3D, ARCGIS PRO, QGIS, TRIMBLE BUSINESS CENTER, LEICA GEO OFFICE, PRIMAVERA P6, MS PROJECT, REVIT
Field Surveying	Total Station, GNSS Receivers (RTK/PPP), Digital Levels, UAV mapping, LiDAR scanning, Hydrographic survey, Cadastral survey
Languages	Sudanese Arabic (native), Old Nubian (native), English (full professional), Chinese (elementary), Indonesian (elementary)

Honours, Awards & Fellowships

Awards

- 2023 **ANSO-CAS-TWAS/UNESCO PhD Scholarship** – Alliance of International Science Organizations & Chinese Academy of Sciences
- 2022 **Master of Engineering with Cum Laude (Distinction)** – Institut Teknologi Sepuluh Nopember (ITS)
- 2019 **KNB Scholarship for Master's Studies** – Indonesian Government (Kemitraan Negara Berkembang)
- 2017 **Academic Excellence Award** – Omdurman Islamic University (highest GPA in Surveying Engineering)
- 2024 **Commendation for Excellent Performance** – UCAS "December 9th Movement" commemorative event

Fellowships

- 2023 – 2027 **ANSO-CAS-TWAS/UNESCO PhD Fellowship** – Chinese Academy of Sciences
- 2020 – 2022 **KNB Indonesian Government Scholarship** – full tuition + living stipend

References

Available upon request.